

6.8 Appendices

Appendix 6.A: Validity of WSSUS in Mobile Radio Channels

The WSSUS assumption greatly simplifies many calculations, and is thus very popular. However, we have to remember that it was developed for ionospheric channels, where a huge number of independent scatterers exist. It is not obvious that this is also a valid assumption for terrestrial wireless channels.¹² In this appendix, we will discuss this validity in more detail. Those considerations might seem somewhat convoluted and abstract at first glance. But it is essential to understand the limitations of the WSSUS concept - there is nothing worse than using a concept only “because everybody else is using it”.

A key question in that context is the definition of expected values. Mathematically, it is clear: we take the expected value over the ensemble. But what is an “ensemble” in a physical channel? If we consider only a certain time instant, then the propagation scenario is completely deterministic, and the impulse response could be obtained in a unique way from a solution of Maxwell’s equations. Similarly, a measured impulse response corresponds to a clearly defined set of measurement parameters (time, location, frequency range....). The ensemble used in the modeling is thus a mathematical construction that inherently makes assumptions about the channel. Most commonly, the impulse responses measured at different times are interpreted as an ensemble. This inherently implies that the channel is ergodic (which is defined as the equivalence of temporal average and ensemble average), with important implications for the validity of WSSUS (see below).

WSS means that the ACF must be independent of the absolute time. It is obvious from physical considerations that the average power (i.e., the ACF for delay zero) decreases when the UE moves away from the BS. Similarly, the statistical properties change when, e.g., an UE moves from a LOS situation into the radio “shadow” of a building that cuts off the LOS. To resolve this problem, Bello introduced the concept of “quasi-stationarity”. A finite time duration is established, during which the stationarity assumption holds; for larger durations (equivalent to larger movements of the UE), stationarity need not hold. This concept is obviously different from the strict mathematical definition of stationarity, as it does not extend over an infinite amount of time. It also requires a good definition of the instantaneous stationarity region - make it too large, and stationarity is not valid; but make it too small, and the number of independent samples obtainable in the region becomes too small. That can be a problem in indoor environments, where the mean power can change over very short distances.

As a further complication, *any* ergodic process is wide-sense stationary. Thus, as soon as we use ergodicity to obtain the ensemble, we implicitly enforce WSS - no matter how large the time is over which we obtain our ensemble (it can be larger than a quasi-stationarity region). The explanation for this seeming paradox lies in the following: when considering a process in the time domain, we can look at “sliding-window” averages (or other statistical properties), and thus see when those properties change significantly. In other words, the fact that two measurements are made within a short time interval tells us something about the stationarity. In an ensemble, this relationship is lost - the way of indexing samples in an ensemble is completely arbitrary.

The US assumption implies that there is no correlation between the fading coming from different scatterers. This is fulfilled in most outdoor environments, but can be violated in indoor environments, especially empty rooms. For example, reflections from the walls are obviously correlated [Kattenbach 1997]. The other interpretation of US is that the correlation between different frequencies depends only on the frequency difference, but not on the absolute frequency. Again, this will be violated if the considered frequencies are outside the stationarity bandwidth, in complete analogy to the stationarity time. It then follows that the number of statistically independent samples that can be obtained within the time/frequency region for which stationarity holds is inherently limited,

¹²Here, “valid” does not mean “fulfilled in an exact mathematical sense”. Rather, it means “leading to an acceptable error for the systems under consideration”. Of course, this definition is ambiguous, as it depends both on the considered system, and on what is deemed “acceptable”.

and all parameters derived from it are affected by an unavoidable inaccuracy due to limited size of the ensemble from which the parameters are estimated.

6.8.2 Appendix 6.B: Instantaneous Channel Parameters

Some papers compute the moments not of the PDP, but of the squared magnitude of an impulse response *realization*. This creates so-called “local” or “instantaneous” parameters. We thus define an “instantaneous PDP”

$$P_h(t, \tau) = |h(t, \tau)|^2 \quad (6.72)$$

The instantaneous mean delay is then

$$T_m(t) = \frac{\int_{-\infty}^{\infty} P_h(t, \tau) \tau d\tau}{\int_{-\infty}^{\infty} P_h(t, \tau) d\tau} \quad (6.73)$$

and the instantaneous rms delay spread is

$$S_\tau(t) = \sqrt{\frac{\int_{-\infty}^{\infty} P_h(t, \tau) \tau^2 d\tau}{\int_{-\infty}^{\infty} P_h(t, \tau) d\tau} - T_m^2(t)} \quad (6.74)$$

The expected value of those instantaneous parameters does *not* agree with the *average* parameters derived from the PDP; for example $E\{S_\tau(t)\} \neq S_\tau$. The question of whether those instantaneous parameters have any physical meaning has been discussed for a long time. A first requirement is that there is a sufficient number of measurements so that a cumulative distribution function of the instantaneous parameters can be taken; just measuring a delay spread at 5 random points in one building has little meaning. Furthermore, instantaneous parameters make only sense if some instantaneous system characteristic, like the bit error probability, can be related to the instantaneous channel parameter. Few such relationships are actually known. For example, the error probability due to delay dispersion is proportional to the *average* delay spread, not the instantaneous one.

It is also important to distinguish here between the instantaneous parameters, which are derived at one particular location/time, and the average parameters in different stationarity regions. In other words, it is possible to describe the cumulative distribution function of the average delay spread in a cell, where each member of the ensemble is derived from an average PDP, i.e., multiple measurements within one stationarity region.